NEW BEDFORD HIGH SCHOOL | SCIENCE DEPARTMENT

Biology Benchmark #1 — Teacher Answer Key and Grading Rubric

Open Response: Population Dynamics | HS-LS2-1 | Total: 10 points

SCORING PRINCIPLES

Accept scientifically accurate equivalent wording. Award evidence and reasoning only when supported by the graph and food web. Score the assessed science content, not spelling or punctuation, unless meaning is unclear.

ANSWER KEY: QUESTIONS 1–3

Q	Pts	Expected answer	Scoring note
1	1	Fruit fly population.	Accept “fruit flies” or equivalent.
2	1	The fruit fly population increases from Day 0 to Day 20 (approximately 230–250 fruit flies by Day 20).	Correct trend word earns the point; exact value is not required.
3	2	Carrying capacity: 300 fruit flies. First reaches it: Day 32.	1 point each for capacity and day.

ANSWER KEY: QUESTION 4

Question 4 **2 pts**	Sample response: The fruit fly population may level off near 300 because predators such as frogs eat fruit flies. The graph shows the population leveling off near 300, and the food-web arrow from fruit fly to frog shows that frogs consume fruit flies. Increased predation can limit further population growth. Accept another accurate limiting factor (resources, space, disease, or competition) when the student explains how it limits fruit flies and uses the visual evidence as directed.

ANSWER KEY: QUESTION 5

Question 5 **4 pts**	Model CER — Claim: If the number of frogs increases, the fruit fly population will likely decrease or level off at a lower number. Evidence: The food web shows an arrow from fruit fly to frog, so frogs eat fruit flies; the graph shows fruit flies leveling off near 300. Reasoning: More frogs means more predation. More fruit flies are eaten, so fewer survive and reproduce, which can lower or limit the population.

BENCHMARK PERFORMANCE LEVELS

Use the 10-point total to report the benchmark performance level below. These performance levels support clear, consistent scoring and reporting across Biology benchmark responses.

Not Meeting 0–3 points 0–39%	Partially Meeting 4–5 points 40–59%	Meeting 6–8 points 60–89%	Exceeding 9–10 points 90–100%
Limited or inaccurate evidence of the standard; response is incomplete or shows major misconceptions.	Some accurate science thinking; response is partial, inconsistent, or missing key evidence/reasoning.	Sufficient, accurate evidence of the standard; response meets the benchmark expectations.	Thorough, accurate, and well-supported evidence of the standard; response is complete and precise.

Teacher scoring workflow: Score Questions 1–5 using the answer key and analytic rubrics; add the points shown below; then record both the total /10 and the corresponding performance level.

Q1	Q2	Q3	Q4	Q5	TOTAL
1 pt	1 pt	2 pts	2 pts	4 pts	10 pts

Consistency checks: Score Q5 elements independently; award a point once when that element is accurate and supported, even if it appears outside its labeled CER section. Do not subtract points for conventions unless meaning is unclear. Use one feedback code to request the next student revision.

Question 5 Analytic Rubric and Feedback Guide

Score each Claim–Evidence–Reasoning element independently. Total question value: 4 points.

QUESTION 5: FOUR-POINT CER RUBRIC

CER element	Pts	Full-credit evidence	No point / incomplete evidence
Claim	1	Predicts that fruit flies decrease or level off at a lower level if frog numbers increase.	Missing, unclear, or conflicts with the food web.
Evidence from graph	1	Accurately notes that the fruit fly population levels off near a carrying capacity of 300.	Does not use the graph or uses it inaccurately.
Evidence from food web	1	Uses fruit fly → frog correctly: frogs consume fruit flies / receive energy from fruit flies.	Does not use the food web, reverses the arrow, or gives an unsupported relationship.
Reasoning	1	Explains: more frogs → more predation → fewer fruit flies survive/reproduce → population lowers or is limited.	Lists evidence without a causal link or gives inaccurate reasoning.

QUESTION 4: TWO-POINT SHORT-RESPONSE RUBRIC

Score	Pts	Descriptor
2	2	Names a valid limiting factor and accurately explains how it could level off the fruit fly population, using the graph and food web as directed.
1	1	Names a valid factor but gives a partial explanation, uses only one visual, or gives a generally correct but incomplete connection.
0	0	Blank, inaccurate, or does not identify a plausible factor and explanation.

RECOMMENDED WRITTEN FEEDBACK CODES

C Clarify the claim or prediction.	**G** Use an accurate graph detail.	**W** Use the food-web relationship; follow the arrow.	**R** Explain how the evidence supports the claim.

Teacher note: For Question 5, feedback should request revision of one visible CER element rather than provide the complete scientific explanation.